

# MS&E 228 (Winter 2026) — Lectures 12 & 13 Notes

## From Causal Graphs to Identification: A Deep Dive

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February 22, 2026

**Readings.** *Applied Causal Inference Powered by ML and AI*, §7, §11. Key papers: Pearl (2009), Richardson & Robins (2013), Cinelli, Forney, & Pearl (2024).

### 1 Beyond Conditional Exogeneity as an Assumption

This chapter marks a pivotal shift in our study of causal inference. For the past several weeks, we have focused on the powerful estimation machinery of causal machine learning, including doubly robust methods and double machine learning. These techniques allowed us to leverage flexible ML models to estimate average and heterogeneous treatment effects, all under a single, unifying assumption: **conditional exogeneity**. We took for granted that, once we control for a sufficiently rich set of covariates  $X$ , the treatment assignment becomes as good as random, i.e.,  $Y(d) \perp\!\!\!\perp D \mid X$ . While this is a powerful and often reasonable starting point, it leaves a critical question unanswered: what does it *mean* for conditional exogeneity to be satisfied?

This is not a mere philosophical question. In practice, we cannot expect to go to a domain expert—a doctor, a sociologist, a business analyst—and ask, “Are the potential outcomes conditionally independent of the treatment assignment given these covariates?” The question is abstract and unintelligible to those who are not steeped in the potential outcomes framework. The goal of this chapter is to introduce a new language for stating our domain assumptions, a language that is both human-interpretable and mathematically rigorous. We want to build a framework where **conditional exogeneity is a conclusion, not an assumption**. It should be the logical result of a set of more primitive, understandable beliefs about how the world works.

To achieve this, we will introduce the framework of **Non-Parametric Structural Equation Models (NP-SEMs)** and their visual representation, **Directed Acyclic Graphs (DAGs)**. This framework will allow us to:

1. **Elicit Domain Knowledge:** Translate qualitative, human-interpretable beliefs about causal relationships into a formal mathematical and graphical structure.
2. **Redefine Potential Outcomes:** Give a new, more intuitive foundation to potential outcomes through the concept of **interventions** on the structural model.

3. **Develop a Formal Test for Identification:** Introduce **d-separation** and **Single World Intervention Graphs (SWIGs)** as a graphical calculus to formally check whether conditional exogeneity holds, and to determine the appropriate set of control variables.

By the end of this chapter, we will have a powerful, principled methodology for moving from a qualitative story about a data-generating process to a rigorous, testable identification strategy. This is the core of causal reasoning.

## 2 Part I: From Domain Knowledge to Causal Graphs

To make these concepts concrete, we will ground our discussion in the hospital messaging case study from the previous chapter. The goal is to understand the causal effect of physicians sending encouraging messages to young patients with diabetes to improve their blood glucose control.

### 2.1 The Hospital Messaging Example: Variables of Interest

Recall the key variables involved in this system:

- **Y:** The patient’s glucose time-in-range in the days following the potential message.
- **D:** The physician’s binary decision to send a message.
- **M:** The actual text content of the message that is sent.
- **X:** The rich set of observable patient data available to the physician in the remote monitoring system (e.g., prior CGM data, activity levels, etc.).

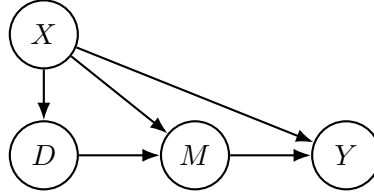
Our previous, high-level argument was that since  $X$  represents all the information the doctor observes, conditioning on  $X$  should be sufficient to satisfy conditional exogeneity. Today, we will scrutinize this argument by building a more detailed model of the world.

### 2.2 Building the Influence Graph: Observed Variables

We begin by creating a graph where each variable is a node and directed edges represent plausible causal influences. This is a collaborative process that should involve domain experts.

1. The patient’s data ( $X$ ) clearly influences the doctor’s decision ( $D$ ), so we draw  $X \rightarrow D$ .
2. The data ( $X$ ) and the decision ( $D$ ) both influence the message content ( $M$ ), so we draw  $X \rightarrow M$  and  $D \rightarrow M$ .
3. The message content ( $M$ ) influences the patient’s outcome ( $Y$ ), so we draw  $M \rightarrow Y$ .
4. The patient’s prior health state ( $X$ ) will also directly influence their future health state ( $Y$ ), so we draw  $X \rightarrow Y$ .

This gives us a basic graph of the observed variables. Notice the assumptions we are already making through the *absence* of edges. For instance, we assume there is no direct arrow from  $D$  to  $Y$ . This encodes the belief that the binary decision itself has no effect on the outcome, except through the message that is sent. The causal effect is entirely mediated by  $M$ .



## 2.3 Adding Unobserved / Latent Variables

Causal inference is fundamentally about what we *don't* see. The next, crucial step is to reason about the unobserved (latent) factors that might influence multiple parts of our system, creating confounding.

- **Unobserved Patient Habits ( $U_p$ ):** Factors like a patient's diet, exercise, or stress levels are not fully captured in the monitoring data  $X$ . These habits will certainly affect both their past health data ( $X$ ) and their future outcomes ( $Y$ ). This introduces a common cause:  $X \leftarrow U_p \rightarrow Y$ .
- **Unobserved Physician Capacity ( $U_c$ ):** Factors like a doctor's workload, time constraints, or even mood are not observed. A busy doctor might be less likely to send a message ( $D$ ) and might write a shorter, less effective message ( $M$ ) when they do. This introduces another common cause:  $D \leftarrow U_c \rightarrow M$ .

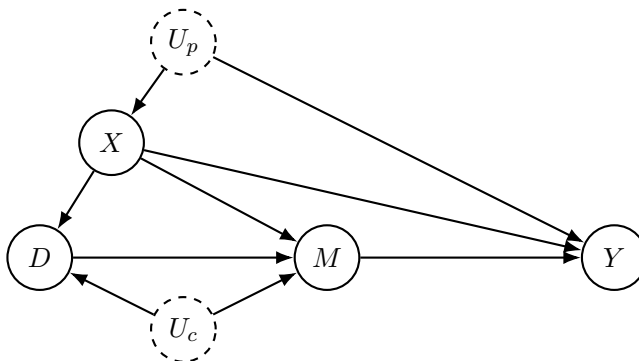
We must also acknowledge that every variable is subject to its own idiosyncratic, independent noise (e.g., measurement error in the CGM, random fluctuations in message wording). We denote these as  $\varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$ . The rule of thumb is: if a latent factor influences only a single observed variable, we can absorb it into that variable's noise term. If it influences two or more variables, it is a potential confounder and must be explicitly added to the graph.

### Can $U_p$ affect $D$ or $M$ ?

An astute reader might ask: could the unobserved patient habits  $U_p$  also influence the doctor's decision  $D$  or the message  $M$ ? The answer depends on the specifics of the system. If the doctor only ever sees the data in  $X$ , then by definition, they cannot react to  $U_p$ . However, if there are other communication channels (e.g., phone calls, prior personal relationships) that reveal information about  $U_p$ , then we would need to add arrows from  $U_p$  to  $D$  and  $M$ . This would make identification much harder. The critical point is that the **omission of these edges is a strong, substantive assumption** that must be justified by domain knowledge. It is in the missing edges that our causal assumptions truly lie.

## 2.4 The Complete DAG and its Meaning

Putting this all together, we arrive at the complete **Directed Acyclic Graph (DAG)**. The solid nodes are observed variables, while the dashed nodes are unobserved (latent) confounders.



This DAG is more than just a picture; it is a formal mathematical object with two key interpretations.

### 2.4.1 The DAG as a Bayesian Network

A DAG is a **Bayesian Network**, which means it implies a specific factorization of the joint probability distribution of all variables. The joint probability is the product of the conditional probability of each node given its parents:

#### Markov Factorization

The probability law of the observed data factorizes according to the graph:

$$p(\mathbf{v}) = \prod_i p(v_i \mid v_{\text{Parents}(i)})$$

In our clinical example this translates to:

$$p(u_p, u_c, x, d, m, y) = p(u_p) p(u_c) p(x \mid u_p) p(d \mid x, u_c) p(m \mid d, x, u_c) p(y \mid m, x, u_p)$$

This factorization is immensely powerful because it encodes all of the conditional independence relationships that must hold in the data. For example, the graph implies that a variable is independent of its non-descendants given its parents. As we will see, the full set of implied independencies can be read directly from the graph using the criterion of d-separation.

### 2.4.2 The DAG as a Generative Model: Non-Parametric Structural Equations

The second, and for our purposes more powerful, interpretation is that the DAG defines a **generative model** for the data.<sup>1</sup> It provides a recipe for how nature constructs the observed world. This

<sup>1</sup>See recent work of [5] and [6] that adopts this lens and connects causal identification with deep generative models, reducing causal identification to gradient descent training.

recipe is formalized through a system of **Non-Parametric Structural Equations (NP-SEMs)**.

The model assumes that every variable in the graph is generated by a function of its direct parents and an independent, exogenous noise term.

**Definition: Non-Parametric Structural Equation Model (NP-SEM)**

An NP-SEM corresponding to a DAG  $\mathcal{G}$  is a set of equations of the form:

$$V_i = f_i(\text{Pa}(V_i), \varepsilon_i)$$

for each variable  $V_i$  in the graph, where:

- $\text{Pa}(V_i)$  are the parents of  $V_i$  in  $\mathcal{G}$ .
- $f_i$  is an unknown, non-parametric function.
- The noise terms  $\varepsilon_i$  are mutually independent across all  $i$ .

For our hospital example, the NP-SEM would be:<sup>2</sup>

$$\begin{aligned} X &= f_X(U_p, \varepsilon_X) \\ D &= f_D(X, U_c, \varepsilon_D) \\ M &= f_M(D, X, U_c, \varepsilon_M) \\ Y &= f_Y(M, X, U_p, \varepsilon_Y) \end{aligned}$$

This system of equations provides a step-by-step procedure for generating a sample from our population. We first draw the exogenous variables (the “root nodes” with no parents, like  $U_p$  and  $U_c$ , and all the  $\varepsilon$  terms) from their respective distributions. You can think of these as random seeds. Then, following the topological order of the graph, we can generate values for all the other variables:

1. Generate  $X$  using  $f_X$  with the drawn values of  $U_p$  and  $\varepsilon_X$ .
2. Generate  $D$  using  $f_D$  with the generated  $X$  and drawn  $U_c, \varepsilon_D$ .
3. Generate  $M$  using  $f_M$  with the generated  $D, X$  and drawn  $U_c, \varepsilon_M$ .
4. Generate  $Y$  using  $f_Y$  with the generated  $M, X$  and drawn  $U_p, \varepsilon_Y$ .

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<sup>2</sup>Formally speaking, we should have also written  $U_p = f_{U_p}(\epsilon_{U_p})$  and  $U_c = f_{U_c}(\epsilon_{U_c})$ , but this is a technicality.

### The Power of What Is NOT Assumed

The term “non-parametric” is crucial. The model imposes no restrictions on the functional forms of the  $f_i$ . They could be linear, nonlinear, highly complex, or simple. The only assumptions are (1) the parental relationships (i.e., the variables that are *not* inputs to each function), and (2) the independence of the noise terms. It is the exclusion of variables from the structural equations—the missing edges in the DAG—that gives the model its power and allows for the possibility of causal identification.

## 3 Part II: Interventions, Potential Outcomes, and SWIGs

We have now established a rich, interpretable model of the *observed* world. But our goal is to estimate a *causal effect* of sending a message. This requires us to reason about counterfactual worlds. The key question is: how do the potential outcomes  $Y(1)$  and  $Y(0)$  relate to the DAG and the NP-SEM we just built?

### 3.1 An Interventional Interpretation of Potential Outcomes

Instead of thinking of potential outcomes as abstract, pre-existing variables that nature reveals to us, the structural framework gives them a more concrete, intuitive meaning. We can define a potential outcome as the outcome we would observe in a hypothetical world where we **intervene** and surgically modify the data-generating process.

- $Y(1)$ : The outcome in a world where we force every physician to send a message (fixing  $D$  to take value 1), regardless of what their natural decision would have been based on  $X$  and  $U_c$ .
- $Y(0)$ : The outcome in a world where we prohibit every physician from sending a message (fixing  $D$  to take value 0).

The average treatment effect,  $\mathbb{E}[Y(1) - Y(0)]$ , is then the difference in the average outcome between these two hypothetical, interventional worlds. We are, in our minds, simulating a perfect randomized controlled trial.

### 3.2 Formalizing Interventions in the SEM

The power of the “structural” in NP-SEM is that the model doesn’t just describe the observed world; it also tells us exactly how to generate data from these interventional worlds. An intervention, denoted  $\text{fix}(D = d)$ , modifies the SEM in a simple way: we ignore the natural value of  $D$  generated by its structural equation and instead use the constant value  $d$ , in every downstream structural equation where  $D$  used to be an input.

## The Interventional Data-Generating Process

To generate data from the interventional world under  $\text{fix}(D = d)$ , we follow these steps:

1. Draw the exogenous noise variables  $(\{\varepsilon_i\}, \{U_j\})$  from the *same* distribution as in the observed world. (The intervention doesn't change patient habits or doctor workload distributions).
2. Generate all variables upstream of  $D$  (including  $D$ ) as usual.
3. Ignore the natural value of  $D$  that was generated and **set**  $D = d$ .
4. Generate all variables downstream of  $D$  using their original structural functions, but plugging in the fixed value  $d$  for  $D$  wherever it appears.

Variables generated in this new world are the potential outcomes, denoted with  $(d)$ .

For our example, the process to generate the potential outcome  $Y(d)$  is:

$$\begin{aligned}
 X &\leftarrow f_X(U_p, \varepsilon_X) \quad (\text{Same as before}) \\
 \text{Natural } D &\leftarrow f_D(X, U_c, \varepsilon_D) \quad (\text{Ignored}) \\
 M(d) &\leftarrow f_M(d, X, U_c, \varepsilon_M) \quad (D \text{ is replaced by the constant } d) \\
 Y(d) &\leftarrow f_Y(M(d), X, U_p, \varepsilon_Y) \quad (\text{Uses the interventional } M(d))
 \end{aligned}$$

This procedure formally defines the potential outcome  $Y(d)$  in terms of the underlying structural model. It also naturally satisfies the **consistency property**: if we were to “intervene” by setting  $D$  to its observed value, we would recover the observed outcome, i.e.,  $Y(D) = Y$ . We have thus given an interventional interpretation of potential outcomes!

### 3.3 The Single World Intervention Graph (SWIG)

We have now come full circle. We want to check conditional exogeneity,  $Y(d) \perp\!\!\!\perp D \mid X$ . We have a graphical tool, the DAG, for checking conditional independencies. However, the original DAG does not contain the nodes  $Y(d)$ . We need a new graph that represents the interventional world and includes both the potential outcome  $Y(d)$  and the natural treatment assignment  $D$  as nodes.

This new graph is the **Single World Intervention Graph (SWIG)**, introduced by Richardson and Robins (2013). The construction is intuitive.

### Constructing the SWIG for $\text{fix}(D=d)$

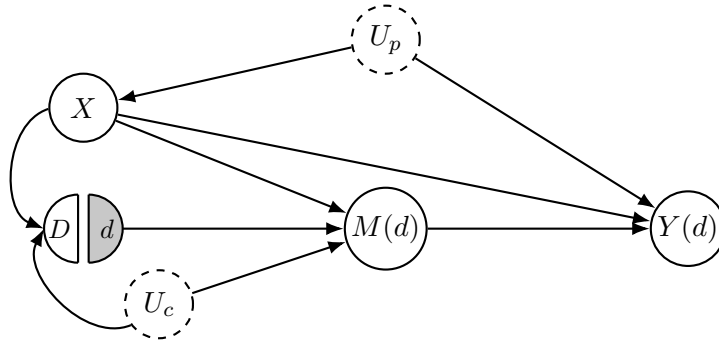
Think of the intervention as a **lightning strike** on the node  $D$ . The node splits into two halves:

- The **left half** represents the *natural* value  $D$  would have taken. All arrows that originally pointed *into*  $D$  now point into this left half.
- The **right half** represents the *fixed, interventional* value  $d$ . All arrows that originally pointed *out of*  $D$  now originate from this right half.

All nodes downstream of the intervention are now potential outcomes (e.g.,  $M$  becomes  $M(d)$ ,  $Y$  becomes  $Y(d)$ ).

The SWIG is a valid Bayesian Network for the joint distribution of variables in a world where we contemplate an intervention but also remember what the natural course of action would have been. Crucially, it contains both the natural treatment assignment  $D$  and the potential outcome  $Y(d)$  as distinct nodes.

For our hospital example, the SWIG for the intervention  $\text{fix}(D = d)$  is:



This brings us to the central insight of this lecture: **checking conditional exogeneity reduces to checking for conditional independence in the SWIG**. The abstract statistical assumption  $Y(d) \perp\!\!\!\perp D \mid X$  is now equivalent to the concrete graphical question: “Is the node for the natural treatment  $D$  separated from the node for the potential outcome  $Y(d)$  in this graph when we condition on the node(s) for  $X$ ?” To answer this, we need a formal graphical criterion for conditional independence in a Bayesian network.

## 4 Part III: d-Separation and Path Analysis

### 4.1 The Graphical Criterion for Independence: d-Separation

We have reduced the problem of checking conditional exogeneity to a problem of checking for conditional independence in a Bayesian Network (the SWIG). The graphical criterion for this is



called **d-separation** (the “d” stands for “directed”). It provides a complete, visual algorithm for determining whether a set of variables  $A$  is independent of another set  $B$ , given a third set  $C$ .

#### Definition: d-Separation

A set of nodes  $C$  **d-separates** a set of nodes  $A$  from a set of nodes  $B$  if and only if  $C$  blocks every undirected path between any node in  $A$  and any node in  $B$ .

An undirected path is a sequence of adjacent nodes, ignoring the direction of the arrows. A path is said to be **blocked** by a set of nodes  $C$  if at least one of the following two conditions holds:

1. The path contains a node  $W$  that is a **non-collider** (a chain  $\rightarrow W \rightarrow$  or a fork  $\leftarrow W \rightarrow$ ) and  $W$  is in the conditioning set  $C$ .
2. The path contains a node  $W$  that is a **collider** ( $\rightarrow W \leftarrow$ ) and neither  $W$  nor any of its descendants are in the conditioning set  $C$ .

If all paths are blocked, then  $A \perp\!\!\!\perp B \mid C$ . If even one path is open, the independence does not hold in general.

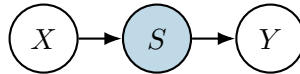
Notice the asymmetric nature of the conditions. For non-colliders, being *in* the conditioning set blocks the flow of information. For colliders, being *out* of the conditioning set blocks the flow; conditioning on them (or their descendants) actively *opens* the path and creates a dependency.

## 4.2 The Three Building Blocks of d-Separation

To build intuition for this seemingly complex definition, we can analyze the three elementary structures that can appear on any path. For each, we will ask: are  $A$  and  $B$  independent given  $C$ ?

### 4.2.1 Building Block 1: The Chain (Mediator)

Consider the path  $A \rightarrow C \rightarrow B$ . Here,  $C$  is a mediator.



The joint distribution factorizes as

$$p(a, c, b) = p(a) p(c|a) p(b|c).$$

### Proof: Conditioning on a Chain Blocks the Path

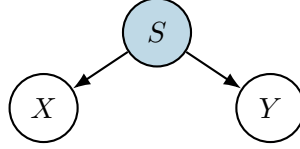
We want to check conditional independence, i.e., check if  $p(a, b|c) = p(a|c)p(b|c)$ .

$$\begin{aligned} p(a, b|c) &= \frac{p(a, b, c)}{p(c)} = \frac{p(a)p(c|a)p(b|c)}{p(c)} \\ &= \frac{p(a, c)p(b|c)}{p(c)} = p(a|c)p(b|c) \end{aligned}$$

Thus,  $A \perp\!\!\!\perp B \mid C$ . Information flows from  $A$  to  $B$  through  $C$ , but once we know the value of the mediator  $C$ , learning about  $A$  provides no additional information about  $B$ . Conditioning on a chain blocks the path.

#### 4.2.2 Building Block 2: The Fork (Common Cause)

Consider the path  $A \leftarrow C \rightarrow B$ . Here,  $C$  is a common cause.



The joint distribution factorizes as

$$p(a, c, b) = p(c)p(a|c)p(b|c).$$

### Proof: Conditioning on a Fork Blocks the Path

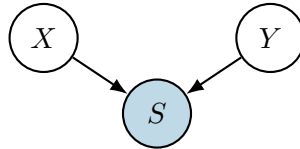
We check if  $p(a, b|c) = p(a|c)p(b|c)$ .

$$p(a, b|c) = \frac{p(a, b, c)}{p(c)} = \frac{p(c)p(a|c)p(b|c)}{p(c)} = p(a|c)p(b|c)$$

Thus,  $A \perp\!\!\!\perp B \mid C$ .  $A$  and  $B$  are correlated because they share a common cause  $C$ . Once we know the value of  $C$ , that common source of variation is explained, and  $A$  and  $B$  become independent. Conditioning on a fork blocks the path.

#### 4.2.3 Building Block 3: The Collider (Common Effect)

Consider the path  $A \rightarrow C \leftarrow B$ . Here,  $C$  is a common effect, or a **collider**.



The joint distribution factorizes as

$$p(a, c, b) = p(a)p(b)p(c|a, b).$$

First, let's check for marginal independence.

$$p(a, b) = \sum_c p(a, b, c) = \sum_c p(a)p(b)p(c|a, b) = p(a)p(b) \sum_c p(c|a, b) = p(a)p(b).$$

So, marginally,  $A \perp\!\!\!\perp B$ . The path is blocked by default when we don't condition on the collider. Now, what happens when we condition on  $C$ ?

#### Proof: Conditioning on a Collider Opens the Path

We check if  $p(a, b|c) = p(a|c)p(b|c)$ .

$$p(a, b|c) = \frac{p(a, b, c)}{p(c)} = \frac{p(a)p(b)p(c|a, b)}{p(c)}$$

There is no reason for this expression to equal  $p(a|c)p(b|c)$ . The term  $p(c|a, b)$  couples  $a$  and  $b$ , and conditioning on  $c$  does not break that coupling. Thus, in general,  $A \not\perp\!\!\!\perp B | C$ . Conditioning on a collider *opens* the path.

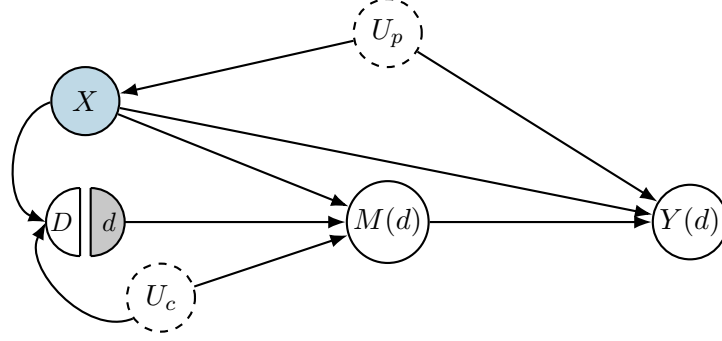
#### Intuition: Selection Bias

The collider is the most counter-intuitive case. The classic example is the relationship between talent ( $A$ ) and luck ( $B$ ) among famous people ( $C = 1$ ). Talent and luck are independent in the general population. However, if we only look at famous people (conditioning on  $C = 1$ ), we introduce a spurious negative correlation. Why? Because to become famous, you need either a lot of talent or a lot of luck (or some combination). If we meet a famous person who we discover has very little talent, we can infer they must have been incredibly lucky. Knowing the value of  $A$  changes our belief about  $B$  once we know  $C$ . This phenomenon is an example of **selection bias**. Conditioning on a common effect creates a dependency between its causes.

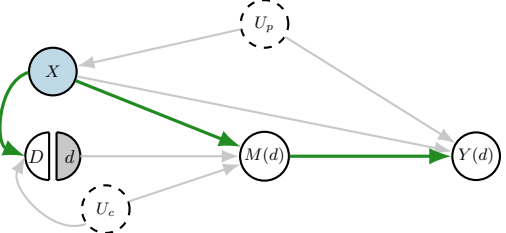
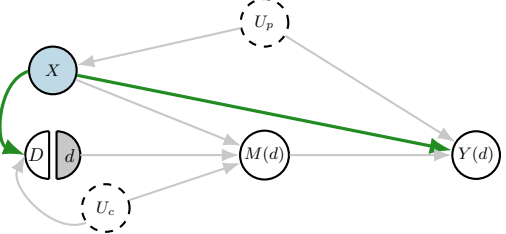
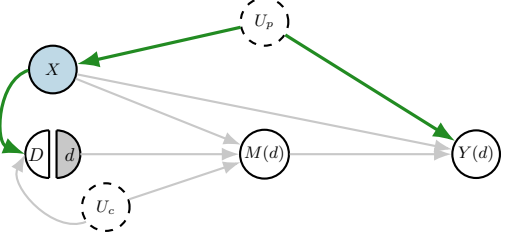
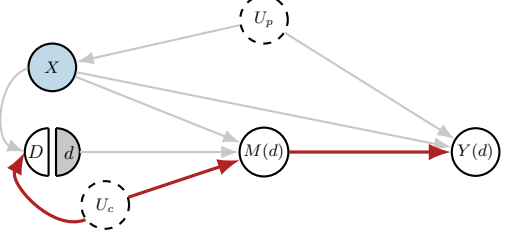
### 4.3 Applying d-Separation to the Hospital Case Study

We are now ready to answer our central question: does conditional exogeneity hold for the intervention  $\text{fix}(D = d)$ ? We must check if  $Y(d) \perp\!\!\!\perp D | X$  in the SWIG. This means we must check if the set  $\{X\}$  d-separates the node  $D$  from the node  $Y(d)$ . We do this by enumerating all undirected paths between  $D$  and  $Y(d)$  and checking if each is blocked by  $\{X\}$ .

Any path from  $D$  to  $Y(d)$  must start by going “backwards” against an arrow into  $D$ . These are often called **backdoor paths** as they represent non-causal associations flowing into the “back door” of the treatment node.



Let's trace the paths:

| Path & Highlighted SWIG                                                                                                                                 | Status         | Reasoning                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Path 1:</b> $D \leftarrow X \rightarrow M(d) \rightarrow Y(d)$      | <b>Blocked</b> | $X$ is a <b>fork</b> ( $\leftarrow X \rightarrow$ ) on this path and is in our conditioning set $\{X\}$ .                                                  |
| <b>Path 2:</b> $D \leftarrow X \rightarrow Y(d)$                     | <b>Blocked</b> | $X$ is a <b>fork</b> ( $\leftarrow X \rightarrow$ ) on this path and is in our conditioning set $\{X\}$ .                                                  |
| <b>Path 3:</b> $D \leftarrow X \leftarrow U_p \rightarrow Y(d)$      | <b>Blocked</b> | $X$ is a <b>chain</b> ( $\leftarrow X \leftarrow$ ) on this path and is in our conditioning set $\{X\}$ .                                                  |
| <b>Path 4:</b> $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$  | <b>Open</b>    | $U_c$ is a <b>fork</b> but is unobserved (not in $\{X\}$ ). $M(d)$ is a <b>chain</b> but is not conditioned on. No colliders exist. The path remains open. |

Since we have found at least one open path, the set  $\{X\}$  does not d-separate  $D$  from  $Y(d)$ .

Therefore, we cannot conclude that **conditional exogeneity holds**. Our initial, high-level intuition was wrong. The mantra of this framework is: **if you cannot verify a property on a SWIG, you should not use it for identification**.

### The Intuition Behind the Failure

What does this open backdoor path mean in plain language? It represents a source of confounding that is not resolved by conditioning on patient data  $X$ . The path shows that the doctor's unobserved capacity,  $U_c$ , influences both their natural decision to send a message ( $D$ ) and the quality of the message they write ( $M$ ).

Imagine two groups of patients who are identical on all observable characteristics  $X$ . In the observational data, the patients who naturally received a message (the  $D = 1$  group) are more likely to have had doctors with high capacity ( $U_c$  is high). These high-capacity doctors not only chose to send a message but also crafted a more thoughtful, effective message ( $M$ ). The patients who did not receive a message (the  $D = 0$  group) are more likely to have had overworked doctors ( $U_c$  is low).

Now, consider the intervention  $\text{fix}(D = 1)$ , where we force *all* doctors to send a message. This includes the overworked doctors. The messages they are forced to write will likely be rushed and less effective than the messages that were sent naturally by the high-capacity doctors. Therefore, the average outcome under the intervention,  $\mathbb{E}[Y(1) \mid X]$ , will be dragged down by these low-quality forced messages, as compare to  $\mathbb{E}[Y \mid D = 1, X]$ . Comparing the observed outcomes of the  $D = 1$  and  $D = 0$  groups, even after conditioning on  $X$ , gives a misleadingly optimistic estimate of the treatment effect, because it conflates the effect of the message with the effect of having a high-capacity doctor.

## 5 Part IV: A Better Intervention and Advanced Topics

The failure of conditional exogeneity for the intervention  $\text{fix}(D = d)$  is a crucial finding. It tells us that simply comparing outcomes for patients who did and did not receive a message, even after controlling for their observable characteristics  $X$ , will lead to a biased estimate of the causal effect. The graphical analysis has not only told us *that* our initial strategy fails, but also *why* it fails: the unobserved confounding from physician capacity  $U_c$ .

This diagnosis points towards two possible solutions:

1. **Measure the Confounder:** We could try to find observable proxies for  $U_c$ . For example, we could collect data on the number of patients a doctor is managing, their time since last break, or other indicators of workload. If we add these proxies to our conditioning set, we might be able to block the backdoor path through  $U_c$ .
2. **Change the Intervention:** Perhaps intervening on the binary decision  $D$  is the wrong question. The analysis revealed that the confounding arises because  $U_c$  affects both  $D$  and

the message content  $M$ . What if we intervened on the message content directly?

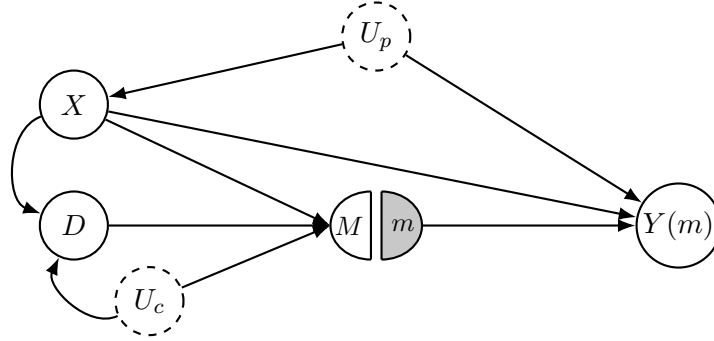
Let's explore the second option, as it leads to a cleaner identification strategy and highlights the power of the framework to guide not just analysis, but also the very definition of the causal question.

### 5.1 A Better Intervention: Standardizing the Message

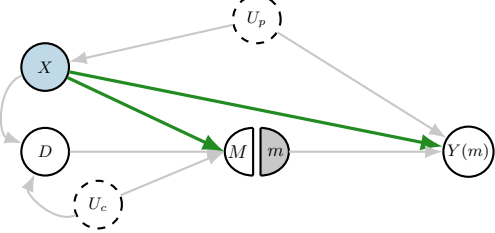
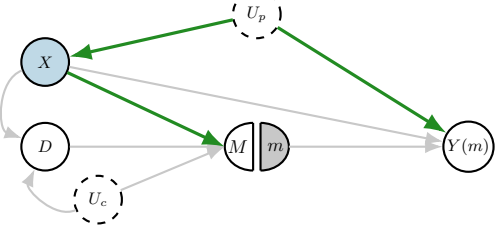
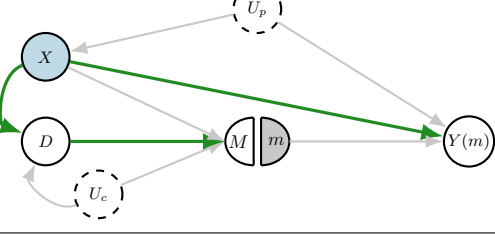
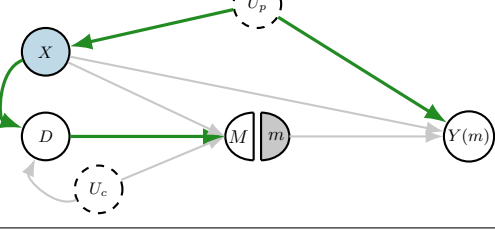
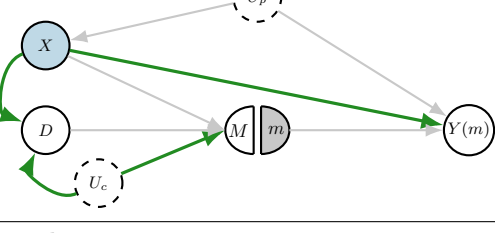
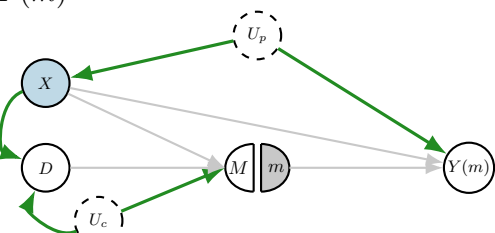
Instead of forcing doctors to send *a* message, what if we intervened to ensure they send a *specific, standardized* message? For example, we could provide all doctors with a high-quality, pre-written template and instruct them to send that exact message. This corresponds to the intervention  $\text{fix}(M = m)$ , where  $m$  is the content of the standardized message.

Our new causal question is: what is the average treatment effect of sending this standardized message,  $\mathbb{E}[Y(m)]$ ? To identify this, we need to check if the corresponding conditional exogeneity assumption holds:  $Y(m) \perp\!\!\!\perp M \mid X$ . That is, among patients with the same observable characteristics  $X$ , is the potential outcome under the standardized message independent of the (potentially empty) message that the patient would have naturally received?

To answer this, we construct a new SWIG for the intervention  $\text{fix}(M = m)$ .



Now, we perform d-separation analysis to check if  $\{X\}$  blocks all paths between the natural message node  $M$  and the potential outcome node  $Y(m)$ .

| Path & Highlighted SWIG                                                                                                                                                          | Status         | Reasoning                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Path 1:</b> $M \leftarrow X \rightarrow Y(m)$                                                | <b>Blocked</b> | $X$ is a <b>fork</b> ( $\leftarrow X \rightarrow$ ) on this path and is in our conditioning set $\{X\}$ .                                                                                                                                                   |
| <b>Path 2:</b> $M \leftarrow X \leftarrow U_p \rightarrow Y(m)$                                 | <b>Blocked</b> | $X$ is a <b>chain</b> ( $\leftarrow X \leftarrow$ ) on this path and is in our conditioning set $\{X\}$ .                                                                                                                                                   |
| <b>Path 3:</b> $M \leftarrow D \leftarrow X \rightarrow Y(m)$                                  | <b>Blocked</b> | $X$ is a <b>fork</b> ( $\leftarrow X \rightarrow$ ) on this path and is in our conditioning set $\{X\}$ .                                                                                                                                                   |
| <b>Path 4:</b> $M \leftarrow D \leftarrow X \leftarrow U_p \rightarrow Y(m)$                  | <b>Blocked</b> | $X$ is a <b>chain</b> ( $\leftarrow X \leftarrow$ ) on this path and is in our conditioning set $\{X\}$ .                                                                                                                                                   |
| <b>Path 5:</b> $M \leftarrow U_c \rightarrow D \leftarrow X \rightarrow Y(m)$                 | <b>Blocked</b> | $D$ is a <b>collider</b> ( $U_c \rightarrow D \leftarrow X$ ) and is <i>not</i> in $\{X\}$ , nor is any descendant of $D$ . The path is blocked by the unconditioned collider. <i>(It is also blocked by <math>X</math> too; both conditions hold here)</i> |
| <b>Path 6:</b> $M \leftarrow U_c \rightarrow D \leftarrow X \leftarrow U_p \rightarrow Y(m)$  | <b>Blocked</b> | $D$ is again a <b>collider</b> ( $U_c \rightarrow D \leftarrow X$ ) and is <i>not</i> in $\{X\}$ . The path is blocked by the unconditioned collider. <i>(It is also blocked by <math>X</math> too; both conditions hold here)</i>                          |

All paths between  $M$  and  $Y(m)$  are blocked by the set  $\{X\}$ . Therefore, we can conclude that **conditional exogeneity holds for this intervention**:  $Y(m) \perp\!\!\!\perp M \mid X$ . This is a powerful result. The graphical analysis has guided us to a different, more robust identification strategy. By changing the causal question from the effect of “sending a message” to the effect of “sending a standardized message,” we have found a setting where identification by conditioning is valid.

### 5.1.1 What We Have Achieved

Let us pause and reflect on what the SWIG framework has accomplished through the hospital messaging example. Starting from interpretable domain assumptions encoded in the DAG, we formalized the data generating process through the non-parametric SEM. We defined interventions and gave an interventional interpretation to potential outcomes and the average treatment effect. We then used the SWIG to visually deduce conditional exogeneity — and discovered that it **fails** for  $\text{fix}(D = d)$  but **holds** for  $\text{fix}(M = m)$ .

The only assumption we elicited from the domain expert was the DAG itself. Conditional exogeneity was a **conclusion** to check, not a premise. And the SWIG revealed a subtle failure that would have been easy to miss with heuristic reasoning alone.

### 5.1.2 The Power of the SWIG: Avoiding Subtle Mistakes

Without the SWIG, we might have naively assumed  $Y(d) \perp\!\!\!\perp D \mid X$  and estimated the effect of “sending a message” by conditioning on patient data  $X$ . This would have been wrong. The SWIG revealed that there is implicit confounding through the mediator  $M$ : the doctor’s workload  $U_c$  affects both the decision to send a message and the quality of the message sent. In the observed data, doctors who chose to send messages typically had more time available, so they sent better, more effective messages. If we force all doctors to send messages under  $\text{fix}(D = 1)$ , those with limited time would send lower-quality messages. Therefore, the observed efficacy of “sending a message” overestimates the true causal effect of the intervention. The fix is to intervene on  $M$  directly (standardize the message content), or additionally condition on physician workload  $U_c$  if it were observable.

## 5.2 Multiple Interventions and the Multiple-Intervention SWIG

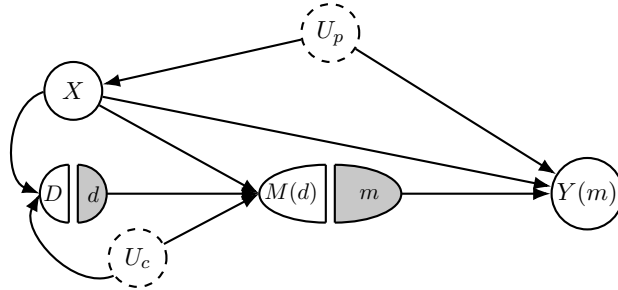
The framework naturally extends to interventions on multiple variables simultaneously. For example, what is the effect of forcing a doctor to send a message *and* forcing that message to have a specific content? This corresponds to the intervention  $\text{fix}(D = d, M = m)$ . The data generating process under this double intervention proceeds as follows: draw all exogenous noise from the same distribution; generate  $X$  and  $D$  as usual; but wherever  $D$  appears as a parent, replace it with the fixed value  $d$ , and wherever  $M$  appears as a parent, replace it with the fixed value  $m$ . The interventional outcome is:

$$Y(d, m) \leftarrow f_Y(m, X, U_p, \varepsilon_Y)$$



A crucial observation is that  $Y(d, m) = Y(m)$ , since the fixed value  $d$  does not directly enter the structural equation for  $Y$ . Once we fix the message content to  $m$ , the binary decision  $D$  no longer matters for the outcome.

We can represent this double intervention in a SWIG where we “strike lightning” on both  $D$  and  $M$ , splitting both variables into their natural and forced values:



However, there is a problem with this SWIG. We would like to check the conditional exogeneity condition that  $Y(m) \perp\!\!\!\perp (D, M) \mid X$ , but  $M$  does not appear as a standalone variable on this graph. Only  $M(d)$  (the left half, representing the natural interventional message; when we force all doctors to send a message) appears. The original, pre-intervention  $M$  is not a variable in this doubly-interventional world. This leads to a key principle that we will also use later on:

#### A Key Principle of SWIGs

If you cannot state a conditional independence using variables that appear on the SWIG, you cannot use d-separation to verify it. And if you cannot verify it on the SWIG, **you should not use it for identification.**

Since we cannot check the required conditional independence on the double-intervention SWIG, we should not use it. The setting here however is special because the potential outcome only depends on  $m$  and not  $d$ . Thus we can try to check the conditional exogeneity condition  $Y(m) \perp\!\!\!\perp (D, M) \mid X$  on the SWIG that corresponds to just  $\text{fix}(M = m)$ . On that SWIG we have both  $Y(m)$  and  $M, D$  appear, and we can verify via  $d$ -separation that this conditional independence property holds. Thus, we can verify this property from that SWIG. More broadly, we should be looking at all SWIGs for which our interventional outcome appears and checking if the conditional independence property holds or not. Typically, this will be a unique SWIG, but sometimes, when the interventional outcome only depends on a subset of the interventions, then we should be considering the SWIG with only that subset of the interventions.

### 5.3 Technical Parenthesis: Primitive Potential Outcomes

As a final technical connection between the potential outcomes framework and the structural equation framework, we can push the idea of multiple interventions to the extreme. What if we intervene on *all* of the variables — fixing every variable’s parents to take some specific values? The interventional potential outcomes that result from this “global intervention” can be thought as the

“primitive potential outcomes”. For our hospital messaging example, these primitive potential outcomes are:

$$\begin{aligned} X(u_p) &\leftarrow f_X(u_p, \varepsilon_X) \\ D(x, u_c) &\leftarrow f_D(x, u_c, \varepsilon_D) \\ M(d, x, u_c) &\leftarrow f_M(d, x, u_c, \varepsilon_M) \\ Y(m, x, u_p) &\leftarrow f_Y(m, x, u_p, \varepsilon_Y) \end{aligned}$$

Each primitive potential outcome is the value a variable would take if all of its parents were exogenously set to specific values.

Because the only thing that remains random is the noise factors, and these noise factors are mutually independent, the resulting **primitive interventional random variables** are also *jointly independent* of each other, i.e. the set of random variables  $\{V_i(v_{\text{parents}(i)})\}_{v_{\text{parents}(i)} \in \mathcal{V}_{\text{parents}(i)}}$  is jointly independent with the set of random variables  $\{V_j(v_{\text{parents}(j)})\}_{v_{\text{parents}(j)} \in \mathcal{V}_{\text{parents}(j)}}$ , where  $\mathcal{V}_{\text{parents}(j)}$  is the set of all possible values that the parents of node  $j$  can take. Moreover, note that because we made no assumption on the form of the structural functions  $f_i$  nor on the form of the noise terms  $\varepsilon_{V_i}$ , any jointly independent distribution on the potential outcomes  $V_i(v_{\text{parents}(i)})$  can be simulated via an appropriately defined NPSEM.

Thus, an equivalent way to present the entire structural equation framework is through these primitives, in a manner closer in spirit to how we defined the potential outcomes framework. Let’s define this alternative way for our hospital example. In **Step 1**, nature generates all primitive potential outcome processes from some exogenous distribution satisfying joint independence:

$$\{Y(m, x, u_p)\} \perp\!\!\!\perp \{M(d, x, u_c)\} \perp\!\!\!\perp \{D(x, u_c)\} \perp\!\!\!\perp \{X(u_p)\} \perp\!\!\!\perp U_p \perp\!\!\!\perp U_c$$

In **Step 2**, we generate the observed data by successive “plugging in” and evaluation: first, we define the observed  $X$  as  $X(U_p)$ ; then we define the observed  $D$  as  $D(X, U_c)$ ; and so on:

$$X \leftarrow X(U_p), \quad D \leftarrow D(X, U_c), \quad M \leftarrow M(D, X, U_c), \quad Y \leftarrow Y(M, X, U_p)$$

Finally, to generate interventional outcomes (e.g., under  $\text{fix}(D = d)$ ), we plug in  $d$  wherever  $D$  appears on the right-hand side:

$$X \leftarrow X(U_p), \quad D \leftarrow D(X, U_c), \quad M(d) \leftarrow M(d, X, U_c), \quad Y(d) \leftarrow Y(M(d), X, U_p)$$

The SEM framework and the potential outcomes framework are thus two sides of the same coin. The SEM gives a **mechanistic** foundation of how the primitive counterfactuals are generated; the potential outcomes framework gives a **counterfactual** foundation. Moreover, typically, we care about average **interventional counterfactuals** (like  $Y(d)$  or  $Y(m)$ ), not averages of these primitive potential outcomes. Hence, the interventional counterfactuals are the most natural analogue

of what people typically define as the potential outcomes in the Neyman–Rubin framework, as opposed to these primitive counterfactuals.

## 6 Part V: A Gallery of Good and Bad Controls

With this rigorous framework in hand, we are now equipped to analyze a gallery of common causal structures to build our intuition about which variables make for “good” or “bad” controls. To solidify our understanding of d-separation and its implications for choosing a valid adjustment set, we will now work through a gallery of common causal structures. For each example, we will be given a DAG representing our domain knowledge and asked to determine which variables, if any, we should condition on to identify the causal effect of a treatment  $D$  on an outcome  $Y$ . These examples are adapted from the excellent “A Crash Course in Good and Bad Controls” by Cinelli, Forney, and Pearl (2024).

### A Note on Practice

When you are faced with a causal inference problem in practice, the first step is always to try to build a DAG. This involves collecting the relevant variables, including potential unobserved confounders, and mapping out their main influences on each other. This is not a solitary exercise; it requires deep conversation with domain experts. In the modern era, Large Language Models (LLMs) can also be a valuable tool in this process. You can query an LLM about other factors that might be at play in a given system, or use it to sanity-check the graph you have constructed. They are useful aids in this crucial first stage of causal inference.

### 6.1 Example 1: Drug Efficacy with Mediation

**Scenario:** A pharmaceutical company wants to estimate the effect of a new drug ( $D$ ) on patient recovery ( $Y$ ). The drug is believed to work by changing the level of a specific biomarker ( $M$ ), which in turn affects recovery. The company has collected data on several variables:

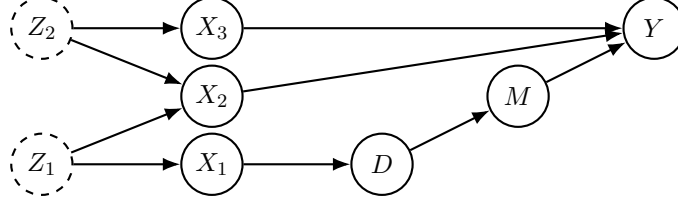
- $D$ : Treatment assignment (taking the drug).
- $Y$ : Patient recovery (the outcome).
- $M$ : Biomarker level (a mediator).
- $X_1$ : Patient’s insurance type, which affects who gets the drug.
- $X_2$ : Disease severity at baseline, which affects recovery.
- $X_3$ : Age-related health decline, which also affects recovery.

Furthermore, domain experts believe there are two key unobserved factors:

- $Z_1$ : Unobserved employer type, which is a common cause of both insurance type ( $X_1$ ) and baseline disease severity ( $X_2$ ).
- $Z_2$ : Unobserved genetic factors or family history, which is a common cause of both baseline disease severity ( $X_2$ ) and age-related decline ( $X_3$ ).

In this first version of the example, we assume that the only factor influencing the treatment decision  $D$  is the patient's insurance type,  $X_1$ .

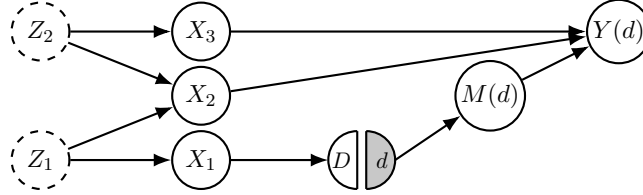
**The DAG:**



**Task:** What is a valid adjustment set to identify the causal effect of  $D$  on  $Y$ ?

**Analysis:** We need to find a set of covariates  $S$  such that  $Y(d) \perp\!\!\!\perp D \mid S$ . We construct the SWIG for  $\text{fix}(D = d)$  and use d-separation to check which sets  $S$  block all backdoor paths from  $D$  to  $Y(d)$ .

**SWIG Analysis for  $\text{fix}(D = d)$ :**



On the SWIG, the non-causal paths from  $D$  to  $Y(d)$  are: (i)  $D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \rightarrow Y(d)$ , and (ii)  $D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \leftarrow Z_2 \rightarrow X_3 \rightarrow Y(d)$ . Every non-causal path must pass through  $X_1$ , which is a chain. Conditioning on  $\{X_1\}$  blocks all such paths, confirming  $Y(d) \perp\!\!\!\perp D \mid X_1$  by d-separation on the SWIG. This is a **good control**. This example also highlights a general principle for a sufficient set of controls: *conditioning on all the parents of the treatment variable is a valid adjustment set*. Any backdoor path must pass through at least one of these parents and the parent can never be a collider on that path, since one of the edges must be going out of the parent.

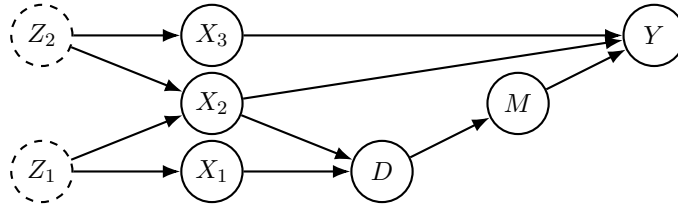
Many other sets are also valid adjustment sets. In particular, anything that contains  $X_1$  and other pre-treatment variables will always be a valid adjustment set. Moreover, another valid adjustment set is  $\{X_2, X_3\}$ .  $X_2$  blocks the path  $D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \rightarrow Y(d)$ , but opens the path  $D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \leftarrow Z_2 \rightarrow X_3 \rightarrow Y(d)$ , on which it is a collider. However,  $X_3$  blocks the latter path.

| Conditioning Set                                     | Valid? | Reasoning                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\{\}$                                               | No     | The path $D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \rightarrow Y(d)$ is open.                                                                                                                                                                                                                              |
| $\{X_1\},$<br>$\{X_1, X_2\},$<br>$\{X_1, X_2, X_3\}$ | Yes    | Every backdoor path must pass through $X_1$ . Since $X_1$ is a chain on every such path, conditioning on $X_1$ blocks all of them. Adding $X_2$ or $X_3$ does not unblock these paths. A crucial rule of thumb: <b>conditioning on all direct parents of the treatment is always a sufficient adjustment set.</b> |
| $\{X_2, X_3\}$                                       | Yes    | $X_2$ blocks the path $D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \rightarrow Y(d)$ , but opens the path $D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \leftarrow Z_2 \rightarrow X_3 \rightarrow Y(d)$ , on which it is a collider. However, $X_3$ blocks the latter path.                               |
| $\{X_2\},$                                           | No     | Conditioning on $X_2$ does not block $D \leftarrow X_1 \leftarrow Z_1$ . Worse, $X_2$ is a collider on $Z_1 \rightarrow X_2 \leftarrow Z_2$ ; conditioning on it opens this path, creating a spurious correlation between $Z_1$ and $Z_2$ .                                                                       |

## 6.2 Example 2: Drug Efficacy with Extra Confounding

**Scenario:** We take the same setup as Example 1, but now we assume that baseline disease severity ( $X_2$ ) also directly influences the doctor's decision to prescribe the drug. This adds the edge  $X_2 \rightarrow D$ .

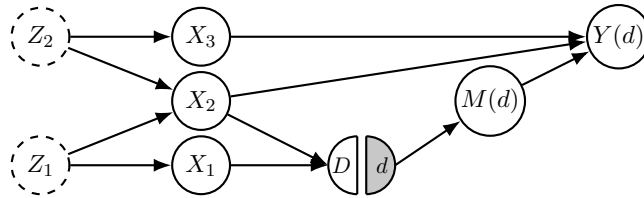
**The DAG:**



**Task:** What is a valid adjustment set now?

**Analysis:** The new edge  $X_2 \rightarrow D$  creates a new backdoor path:  $D \leftarrow X_2 \rightarrow Y(d)$ . This is a classic confounding fork. To achieve identification, we must block this new path *in addition* to the paths that existed before.

**SWIG Analysis for  $\text{fix}(D = d)$ :**



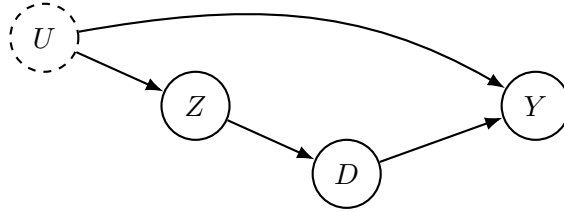
On the SWIG, the new edge  $X_2 \rightarrow D$  creates additional non-causal paths. Consider conditioning on  $\{X_2\}$  alone: while it blocks  $D \leftarrow X_2 \rightarrow Y(d)$ , the node  $X_2$  is a collider on the segment  $Z_1 \rightarrow X_2 \leftarrow Z_2$ , so conditioning on it opens the path  $D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \leftarrow Z_2 \rightarrow X_3 \rightarrow Y(d)$ . Conditioning on  $\{X_1, X_2\}$  blocks all non-causal paths:  $X_2$  blocks the fork at  $X_2$ , and  $X_1$  blocks all paths through  $X_1$ . Thus  $Y(d) \perp\!\!\!\perp D \mid X_1, X_2$  holds by d-separation on the SWIG. Both  $X_1$  and  $X_2$  are **good controls**. Similarly,  $\{X_2, X_3\}$  is a valid adjustment set, since  $X_2$  blocks the fork at  $X_2$ , while  $X_3$  blocks the path through the  $Z$ 's, on which  $X_2$  is a collider.

| Conditioning Set                                            | Valid? | Reasoning                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\{X_1\}$                                                   | No     | It blocks the path through $X_1$ , but leaves the new path $D \leftarrow X_2 \rightarrow Y(d)$ wide open.                                                                                                                                                                                                        |
| $\{X_2\}$                                                   | No     | While it blocks the new path through $X_2$ , it leaves the original path through the $Z$ 's open: $D \leftarrow X_1 \leftarrow Z_1 \rightarrow X_2 \leftarrow Z_2 \rightarrow X_3 \rightarrow Y(d)$ . Conditioning on $X_2$ does <i>not</i> block this path at $X_2$ , because $X_2$ is a collider on this path. |
| $\{X_1, X_2\}, \quad \{X_2, X_3\}, \quad \{X_1, X_2, X_3\}$ | Yes    | This set blocks both types of backdoor paths.                                                                                                                                                                                                                                                                    |

### 6.3 Example 3: Complete Proxy of an Unobserved Confounder

**Scenario:** We want to estimate the effect of prenatal multivitamin consumption ( $D$ ) on the risk of birth defects ( $Y$ ). We suspect there are unobserved genetic factors ( $U$ ) that are a common cause of both the mother's likelihood of taking vitamins and the child's risk of birth defects ( $U \rightarrow D$  and  $U \rightarrow Y$ ).  $U$  is unobserved. However, we do observe the family's medical history ( $Z$ ), which is influenced by the genetic factor ( $U \rightarrow Z$ ). The mother's decision to take vitamins is driven by this observed family history ( $Z \rightarrow D$ ), not directly by the unobserved genetics, i.e., there is no direct edge  $U \rightarrow D$ .

**The DAG:**

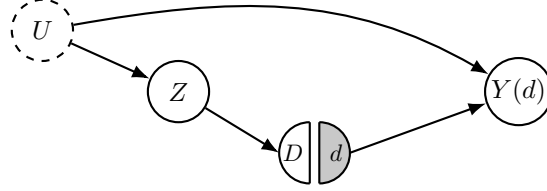


**Task:** Should we condition on the family history,  $Z$ ?

**Analysis:** Yes. The confounding path is  $D \leftarrow Z \leftarrow U \rightarrow Y(d)$ .  $Z$  is a chain on this path. By conditioning on  $Z$ , we block the flow of association from the unobserved confounder  $U$  to the treatment  $D$ . Even though we cannot observe  $U$  itself, we can neutralize its confounding effect by conditioning on its proxy,  $Z$ .  $Z$  is a **complete treatment proxy** because it fully mediates

the effect of  $U$  on  $D$ . This is a powerful and common strategy when the true confounder is not measurable.

**SWIG Analysis for  $\text{fix}(D = d)$ :**

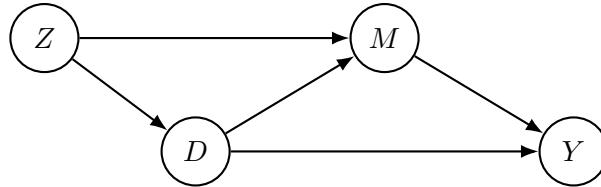


On the SWIG, the only non-causal path is  $D \leftarrow Z \leftarrow U \rightarrow Y(d)$ . The node  $Z$  is a chain on this path. Conditioning on  $Z$  blocks it, establishing  $Y(d) \perp\!\!\!\perp D \mid Z$  by d-separation on the SWIG.  $Z$  is a **good control**: even though the true confounder  $U$  is unobserved, its complete proxy  $Z$  suffices.

#### 6.4 Example 4: Treatment-Mediator Confounder

**Scenario:** We are estimating the effect of an intensive exercise program ( $D$ ) on long-term cardiovascular health ( $Y$ ). The program is believed to work partly by encouraging participants to adopt a healthier diet ( $M$ ), which then improves health ( $D \rightarrow M \rightarrow Y$ ). We observe socioeconomic status ( $Z$ ), which is a common cause of both enrollment in the expensive exercise program and baseline diet quality ( $Z \rightarrow D$  and  $Z \rightarrow M$ ).

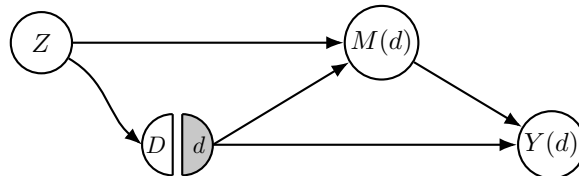
**The DAG:**



**Task:** Should we condition on socioeconomic status,  $Z$ ?

**Analysis:** Yes. This structure is very similar to our original hospital messaging case study. There is a backdoor path  $D \leftarrow Z \rightarrow M(d) \rightarrow Y(d)$ . On this path,  $Z$  is a fork. Conditioning on  $Z$  blocks this path. This highlights a critical lesson: **do not forget confounders of the mediator**. A variable that doesn't directly cause the outcome  $Y$  can still be a crucial confounder if it affects both the treatment  $D$  and a downstream mediator  $M$ . Failing to control for it will lead to biased estimates.

**SWIG Analysis for  $\text{fix}(D = d)$ :**



On the SWIG, the non-causal path is  $D \leftarrow Z \rightarrow M(d) \rightarrow Y(d)$ . The node  $Z$  is a fork on this path. Conditioning on  $Z$  blocks it, establishing  $Y(d) \perp\!\!\!\perp D \mid Z$  by d-separation on the SWIG.  $Z$  is a **good control**: do not forget confounders of the mediator.

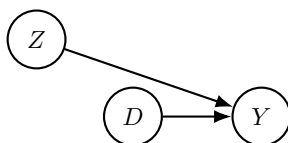
## 6.5 Example 5: Neutral Controls

Some variables are “neutral” in the sense that conditioning on them is not required for identification, but may be beneficial or detrimental for statistical precision.

### 6.5.1 Case 5a: Outcome-Only Cause

**Scenario:** We are studying the effect of a tutoring program ( $D$ ) on student test scores ( $Y$ ). We also observe the student’s IQ ( $Z$ ), which we assume affects their test score but not whether they receive tutoring.

**The DAG:**

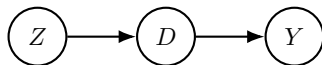


**Analysis:** There are no backdoor paths between  $D$  and  $Y$ . Therefore, the empty set  $\{\}$  is a valid adjustment set; we have unconditional exogeneity,  $Y(d) \perp\!\!\!\perp D$ . Conditioning on  $Z$  is not necessary to remove bias. However, since  $Z$  is a strong predictor of the outcome  $Y$ , conditioning on it will explain away a large portion of the variance in  $Y$ , leading to a more precise estimate of the treatment effect. In this case,  $Z$  is a **good control** to include for statistical reasons, even if not for identification.

### 6.5.2 Case 5b: Treatment-Only Cause

**Scenario:** We are again studying the tutoring program ( $D$ ) and test scores ( $Y$ ). This time, we observe the student’s distance from the tutoring center ( $Z$ ), which affects who enrolls in tutoring but has no direct effect on their test score.

**The DAG:**



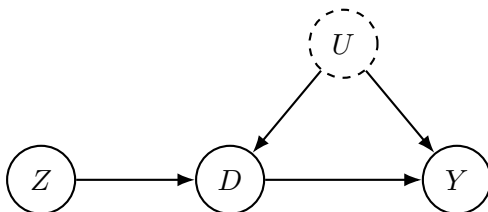
**Analysis:** Again, there are no backdoor paths. The empty set is a valid adjustment set. Should we condition on  $Z$ ? No. Conditioning on a variable that only predicts the treatment (an “instrumental variable”) is generally a bad idea when using a conditioning-based estimator. It removes “good” variation from the treatment that is not associated with any confounding, which can increase the variance of our estimate. In the extreme, if  $Z$  were a perfect predictor of  $D$ , conditioning on it would leave no variation in  $D$  to analyze, destroying any hope of estimation. Here,  $Z$  is a **bad control** from a precision standpoint.



## 6.6 Example 6: The Instrument (A Bad Control in This Context)

**Scenario:** We want to estimate the effect of military service ( $D$ ) on lifetime earnings ( $Y$ ). There is unobserved confounding from family background ( $U$ ), which affects both the likelihood of serving and future earnings ( $U \rightarrow D$  and  $U \rightarrow Y$ ). We also observe a person's draft lottery number ( $Z$ ), which was randomly assigned and strongly affects only the likelihood of serving ( $Z \rightarrow D$ ).

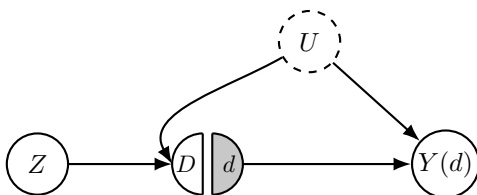
**The DAG:**



**Task:** Should we condition on the lottery number,  $Z$ ?

**Analysis:** No. This is a classic example of an **Instrumental Variable (IV)**, which we will study in detail next week. For now, let's analyze it as a control.

**SWIG Analysis for  $\text{fix}(D = d)$ :**

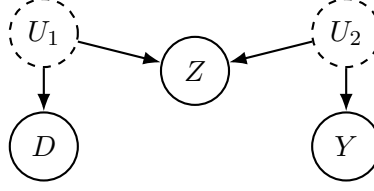


The confounding path is  $D \leftarrow U \rightarrow Y(d)$ . Conditioning on  $Z$  does nothing to block this path. In fact, it makes things worse. By conditioning on the random part of the treatment assignment ( $Z$ ), the only remaining variation in  $D$  is the part driven by the unobserved confounder  $U$ . This amplifies the bias. Conditioning on an instrument is a terrible idea when there is unobserved confounding and you are using an estimator that assumes conditional exogeneity.

## 6.7 Example 7: Pre-Treatment Collider (M-Bias)

**Scenario:** We are studying peer effects in civic engagement. We want to know if Friend 1's engagement ( $D$ ) affects Friend 2's engagement ( $Y$ ). We assume there is **homophily**: people with similar unobserved personality traits tend to become friends. Let  $U_1$  be the traits of Friend 1, and  $U_2$  be the traits of Friend 2. These traits affect their respective engagement levels ( $U_1 \rightarrow D$ ,  $U_2 \rightarrow Y$ ) and also the likelihood that they form a friendship ( $Z$ ) in the first place ( $U_1 \rightarrow Z$ ,  $U_2 \rightarrow Z$ ).

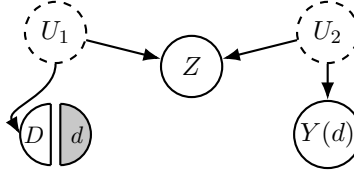
**The DAG:**



**Task:** A common strategy in network analysis is to restrict the analysis to pairs who are actually friends (i.e., condition on  $Z = 1$ ). Is this a good idea?

**Analysis:** No. This is a classic case of **M-bias**.

**SWIG Analysis for  $\text{fix}(D = d)$ :**



On the SWIG, the only path connecting  $D$  and  $Y(d)$  is  $D \leftarrow U_1 \rightarrow Z \leftarrow U_2 \rightarrow Y(d)$ . Without conditioning, this path is blocked at the collider  $Z$ . Conditioning on  $Z$  opens the collider, creating a spurious association between  $D$  and  $Y(d)$  through the unobserved traits  $U_1$  and  $U_2$ . The d-separation analysis on the SWIG confirms that the empty set is valid ( $Y(d) \perp\!\!\!\perp D$ ) and that conditioning on  $Z$  destroys this independence.  $Z$  is a **bad control**.

This is a very interesting example:  $Z$  is pre-treatment variable, i.e., its value is determined prior to the value of the treatment variable, in the arrow-of-time. However, conditioning on it is worse than not conditioning on it! This example showcases that the typical empirical practice of “control for all pre-treatment variables”, can actually go wrong (even though in most practical settings it is a reasonable approach).

## 6.8 Example 8: Conditioning on a Mediator

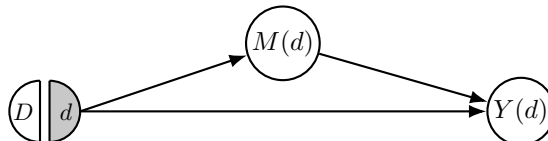
**Scenario:** We are estimating the total effect of a job training program ( $D$ ) on wages ( $Y$ ). The program works by improving skills ( $M$ ), which then increase wages ( $D \rightarrow M \rightarrow Y$ ). There may also be a direct effect of the program on wages, e.g., through certification ( $D \rightarrow Y$ ).

**The DAG:**  $D \rightarrow M \rightarrow Y$  and  $D \rightarrow Y$ .

**Task:** Should we condition on the post-treatment variable  $M$ ?

**Analysis:** No, not if we want the total effect.  $M$  is a mediator; it lies on the causal pathway from  $D$  to  $Y$ . Conditioning on  $M$  blocks the flow of causality through that path.  $M$  is a **bad control** for the total effect.

**SWIG Analysis for  $\text{fix}(D = d)$ :**



On the SWIG for  $\text{fix}(D = d)$ , the variable  $M$  does not appear—only its interventional version  $M(d)$  does. The conditional independence statement  $Y(d) \perp\!\!\!\perp D \mid M$  cannot even be stated using variables on the SWIG, because  $M$  is not a variable in this interventional world. Since we cannot verify it on the SWIG, **it should not be used for identification**.

It can be shown that in this setting, what we would estimate is the **Controlled Direct Effect (CDE)**: the effect of the program that does *not* operate through skills.

Controlling for  $M$  estimates the “Controlled Direct Effect”

Controlling for  $M$  and applying identification by conditioning, we are estimating a form of a “direct effect”:

$$\text{Controlled-Direct-Effect (CDE)} = \mathbb{E}[Y(1, M) - Y(0, M)]$$

While the controlled direct effect might be an interesting question in its own right, it is not the total causal effect. The fact that controlling for  $M$  estimates the CDE would also be the case if there was an unobserved confounder between  $D$  and  $M$ , i.e.  $D \leftarrow U \rightarrow M$ . However, as we will see in the next section this is no longer the case if there is an unobserved confounder between  $Y$  and  $M$ .

**Interpretation of the CDE.** The CDE can be explained by the following thought process: Intervene on both  $D$  and  $M$ , i.e.,  $\text{fix}(D = d, M = m)$ . Look at differences when we flip the treatment variable, while mediator stays the same, i.e.,

$$\text{Individual-Direct-Effect}(m) = Y(1, m) - Y(0, m).$$

Then take the average of this individual direct effect, over the randomness in the potential outcomes and over the values of  $m$ , drawn from the non-interventional distribution of  $M$ :

$$\text{CDE} = \mathbb{E}[\text{Individual-Direct-Effect}(M)].$$

Other notions of direct and indirect effects have been proposed in the literature on *mediation analysis*. These various notions basically differ in the distribution that we use to average over the values of the mediator  $m$ . For instance, the natural direct effect uses the distribution of mediators  $M(0)$  under the control intervention:

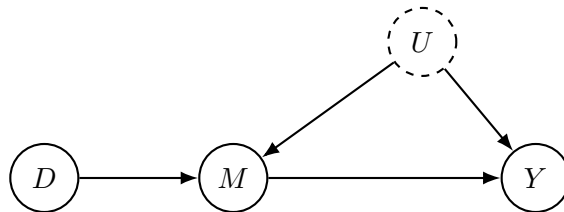
$$\text{Natural-Direct-Effect (NDE)} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))]$$

Controlling for  $M$  does not, generally, estimate the Natural Direct Effect but only the Controlled Direct Effect.

## 6.9 Examples 9: The Confounded Mediator and the Birthweight Paradox

**Scenario:** A new drug ( $D$ ) is believed to affect survival ( $Y$ ) only by changing a biomarker ( $M$ ). However, there is an unobserved genetic factor ( $U$ ) that is a common cause of both the biomarker level and survival ( $U \rightarrow M$  and  $U \rightarrow Y$ ).

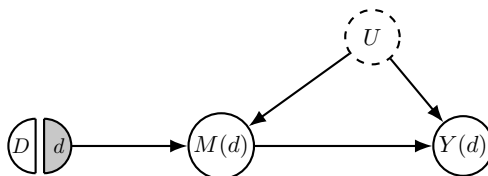
**The DAG:**



**Task:** Should we condition on the biomarker  $M$ ?

**Analysis:** No. This is even worse than the simple mediator case. Here,  $M$  is a **confounded mediator**. It is also a collider on the path  $D \rightarrow M \leftarrow U \rightarrow Y$ . If we condition on  $M$ , we open this path, creating a spurious association between  $D$  and  $Y$  through the unobserved confounder  $U$ . We would estimate a non-zero effect even if the true total effect were zero. This is the structure underlying the famous **birthweight paradox**, where maternal smoking ( $D$ ) appears protective for low-birthweight babies ( $M = 1$ ) because of unobserved birth defects ( $U$ ) that cause both low birthweight and infant mortality ( $Y$ ). This is strictly worse than Example 8: not only does conditioning on  $M$  fail to estimate the total effect, it does not even estimate the Controlled Direct Effect.  $M$  is a **bad control**.

**SWIG Analysis for  $\text{fix}(D = d)$ :**

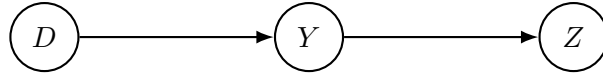


As with Example 8, the variable  $M$  does not appear on the SWIG—only  $M(d)$  does. The conditional independence  $Y(d) \perp\!\!\!\perp D \mid M$  cannot be stated using variables on the SWIG, so it cannot be verified by d-separation and should not be used for identification.

## 6.10 Example 11: Conditioning on an Outcome of the Outcome

**Scenario:** A study in the 19th century studied the effect of improvement in nutrition quality ( $D$ ) on height ( $Y$ ) and found that it led to shorter people. However, the study was conducted solely using men that had enlisted in the army or where in prison  $Z$ . Moreover, enlisting in the army can be affected by a person's height (taller men had more civilian opportunities and were less likely to enlist in the army).

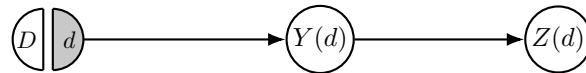
**The DAG:**



**Task:** Looking at only this particular sub-population means that we are conditioning on  $Z$ . Is this a good control?

**Analysis:** No. Conditioning on a variable that is downstream of the outcome is a form of selection bias. It can distort the relationship between  $D$  and  $Y$  in unpredictable ways. It is a bad control.

**SWIG Analysis for  $\text{fix}(D = d)$ :**



On the SWIG, the variable  $Z$  does not appear—only its interventional version  $Z(d)$  does. The conditional independence  $Y(d) \perp\!\!\!\perp D \mid Z$  cannot be stated using variables on the SWIG, so it cannot be verified by d-separation and should not be used for identification. Conditioning on a downstream outcome is a form of selection bias.  $Z$  is a **bad control**.

## 7 Summary and Takeaways

### 7.1 Summary of Good and Bad Controls

The gallery of examples illustrates that the decision to include a variable in a regression model is not statistically innocent; it is a causal decision. The effect of conditioning depends entirely on the variable’s role in the underlying causal graph.

| Node Type           | Structure                   | Conditioning           |
|---------------------|-----------------------------|------------------------|
| Collider            | $\rightarrow V \leftarrow$  | <b>Opens</b> the path  |
| Common cause (fork) | $\leftarrow V \rightarrow$  | <b>Blocks</b> the path |
| Mediator (chain)    | $\rightarrow V \rightarrow$ | <b>Blocks</b> the path |

#### High-Level Rules of Thumb

- **Pre-treatment:** Adjust only for variables that are **ancestors** of  $D$ ,  $Y$ , or both. Avoid colliders and instruments (especially for confounded treatments via unobserved confounders).
- **Post-treatment:** Almost always a **bad idea** to adjust. Avoid mediators, colliders, and outcomes of the outcome. Watch out for controlling for mediators, that are confounded with the outcome, even when you care about the direct effect of the treatment and not the total effect.

## 7.2 The General Framework: A Recap

This chapter has introduced a complete, rigorous framework for moving from qualitative domain knowledge to a quantitative identification strategy.

1. We start by building a **DAG** that represents our assumptions about the causal relationships between all relevant variables, both observed and unobserved.
2. This DAG corresponds to a **Non-Parametric Structural Equation Model (NP-SEM)**, a formal generative model of the world.
3. We define the causal effect of interest as an **intervention** on this model, which gives rise to potential outcomes.
4. We construct the **Single World Intervention Graph (SWIG)** to represent the interventional world.
5. We use the graphical criterion of **d-separation** on the SWIG to formally check if a proposed identification strategy, such as conditional exogeneity, is valid.

## 7.3 Takeaways

1. **d-Separation is the key tool** for checking conditional independence in a DAG. The ability to trace paths and check for blockers is the fundamental skill of graphical causal inference.
2. **Not all controls are good controls.** The decision to condition on a variable must be justified by its position in the causal graph, not by statistical correlations alone.
3. **Beware of colliders.** Conditioning on common effects, or variables affected by them, is a primary source of bias in observational studies (selection bias).
4. **The SWIG is a formal verification tool.** It transforms vague concerns about confounding into a precise, checkable graphical statement. Always verify your identification strategy on the SWIG before proceeding to estimation.

This framework provides the bedrock for all of causal inference. In the coming weeks, we will see how it can be extended to derive other powerful identification strategies beyond simple identification by conditioning, such as instrumental variables and dynamic treatment regimes.

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